Harish Baki

Postdoctoral Associate/ML for Weather & Climate

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PROFESSONAL SUMMARY

Atmospheric scientist and machine learning researcher specializing in building and improving probabilistic weather forecast systems for extreme wind, precipitation, and renewable energy applications. Experienced in numerical weather prediction calibration, statistical post-processing, ensemble evaluation, and deep learning–based downscaling at grid-relevant spatial scales. Strong Python scientific computing background with extensive experience managing large, high-resolution atmospheric datasets using Xarray, Dask, and Zarr in multi-GPU HPC environments. Proven ability to translate atmospheric dynamics into operationally relevant wind and ramp risk insights.

FORECAST SYSTEM DEVELOPMENT & UNCERTAINTY MODELING

- Developed and improved WRF forecast systems through sensitivity analysis, parameter calibration, and data assimilation to enhance tropical cyclone prediction skill.
- Designed deep learning frameworks for wind gust nowcasting and high-resolution wind field reconstruction from sparse Mesonet observations.
- Built attention-based U-Net architectures for precipitation forecasting under spatiotemporal shifts (NeurIPS Weather4Cast, 4th place).
- Lead development of a multi-model deep learning ensemble for downscaling global and regional climate products to 2.5 km resolution using U-Net variants, attention-based transformers, and GANs, integrating ERA5, CONUS404, EDDEv2, URMA, and ICON datasets.
- Designed ensemble-based evaluation strategies to assess calibration, spatial coherence, extreme event representation, and ramp behavior.
- Applied Gaussian Process Regression and Bayesian optimization for uncertainty quantification and model calibration.

EDUCATION

Indian Institute of Technology Madras — Chennai, Tamil Nadu, India

Integrated MS & Ph.D., Mechanical Engineering — 2017-2022

- Dissertation: "Sensitivity based Calibration Strategy with Data Assimilation to Improve the Prediction of Cyclones over the Indian Subcontinent"

Rajiv Gandhi University of Technology – NUZVID — Nuzivid, Andhra Pradesh, India

Bachelor of Technology (Honors), Mechanical Engineering — 2011-2015

PROFESSIONAL EXPERIENCE

University at Albany — Albany, New York, the US

Postdoctoral Associate — September 2024 - present

Atmospheric Sciences Research Center

- Lead development of deep learning ensemble for 2.5 km downscaling of future climate and reanalysis products.

- Built gust nowcasting system from sparse Mesonet observations using PyTorch-based architectures.
- Designed benchmarking pipelines to evaluate extreme wind and ramp prediction performance.

Technical University of Delft Delft, The Netherlands

Postdoctoral Researcher September 2022 -August 2024

Department of Geosciences and Remote Sensing

Faculty of Civil Engineering and Geosciences

- Led development of high-resolution (500 m) offshore wind and solar resource assessments using the Weather Research and Forecasting (WRF) model to generate grid-relevant wind fields for renewable energy planning.
- Designed and implemented a machine learning-based wind profile reconstruction framework from global reanalysis datasets using TabNet, improving vertical wind characterization for turbine-relevant heights.
- Analyzed frontal low-level jets and extreme wind ramp events, quantifying their impact on wind power variability and operational ramp risk.
- Conducted validation and benchmarking of model outputs against reanalysis and observational datasets to assess forecast reliability and extreme-event representation.

Indian Institute of Technology Madras Chennai, Tamil Nadu, India

Institute Postdoctoral Equivalent Fellowship January 2022 – June 2022

- Co-led development and calibration of a coupled COAWST–WRF air–sea interaction forecasting system, delivered to the Indian Space Research Organization (ISRO) for operational evaluation.

RESEARCH INTERESTS

Numerical weather prediction; atmospheric boundary layer processes; machine learning for weather and climate; extreme weather and hydrometeorological prediction; deep learning for wind gust nowcasting and quantitative precipitation forecasting; statistical post-processing of NWP outputs; reanalysis and observational data integration (ERA5, CERRA, lidar profilers, Mesonet observations).

SELECTED ACCOMPLISHMENTS

- Fourth place of the NeurIPS-2023 competition: 2023
  https://weather4cast.net/neurips2023

GRANTS

- NWO/EINF-grant EINF-8612 2024
  Awarded 1 million standard billing units (SBUs) for the proposal titled "European Scalable Complementary Offshore Renewable Energy Sources (EU-SCORES) - Solar-Wind hindcasts (2024)"

MACHINE LEARNING ACTIVITIES

Participation in Competitions

- NeurIPS-2023 Weather4Cast: Super-Resolution Rain Movie Prediction under Spatiotemporal Shifts 2023
  Ranked 4th, https://weather4cast.net/neurips2023

Online Courses

- Coursera: Introduction to Machine learning 2017

TECHNICAL SKILLS IN MACHINE LEARNING

Programming & Data Engineering

- Python (NumPy, Pandas, SciPy), Xarray, Dask, Zarr

Machine Learning & Forecast Modeling

- PyTorch, TensorFlow, Torch Lightning
- Gaussian Process Regression, Bayesian Optimization
- CNNs, U-Net, Swin Transformer, GANs
- Statistical post-processing & calibration

High-Performance Computing

- Multi-GPU distributed training, NVIDIA A100 environments, large-scale climate data processing

Numerical Weather Prediction

- WRF calibration, sensitivity analysis, data assimilation
- Reanalysis & regional climate products (ERA5, CERRA, ICON, CONUS404, EDDEv2, URMA)

JOURNAL PUBLICATIONS

Google Scholar Citations: https://scholar.google.com/citations?user=2q-QsUIAAAAJ&hl=en&authuser=1
ResearchGate Page: https://www.researchgate.net/profile/Harish-Baki
ORCID: 0000-0003-1956-8280
SCOPUS: 57226340925

[15] **Baki, H.,** & Basu, S., (2026). A Deep Learning Framework for Nowcasting Wind Gusts from Sparse Observations via Gridded synthetic Initializations. *Journal of Advances in Modeling Earth Systems*. (Under preparation).

[14] **Baki, H.,** & Basu, S., (2026). A Deep Learning Framework for Generating High-Resolution Wind Gust Fields from Sparse Mesonet Observations. *Journal of Artificial Intelligence for the Earth Systems* (revision submitted). Preprint available at 10.22541/essoar.176031304.41826798/v2.

[13] **Baki, H.,** & Basu, S., (2026). A Chebyshev Polynomial-based Wind Speed Profile Characterization Framework: Applications in Mesoscale Model Evaluation. *Wind Energy.* **29, no. 2 (2026): e70080**, https://doi.org/10.1002/we.70080.

[12] Lavidas, G., Mezilis, L., Alday, M., **Baki, H.,** Tan, J., Jain, A., Engelfried, T. and Raghavan, V., (2025). Marine renewables in Energy Systems: Impacts of climate data, generators, energy policies, opportunities, and untapped potential for 100% decarbonised systems. *Energy*, **p.138359**. https://doi.org/10.1016/j.energy.2025.138359

[11] **Baki, H.,** Basu, S., & Lavidas, G., (2025). Modelling Frontal Low-Level Jets and Associated Extreme Wind Power Ramps over the North Sea. *Wind Energy Science*, **10, 1575–1609**. DOI: 10.5194/wes-10-1575-2025

[10] **Baki, H.,** Basu, S., & Lavidas, G., (2024). Estimating the Offshore Wind Power Potential of Portugal by Utilizing Gray-Zone Atmospheric Modeling. *Journal of Renewable and Sustainable Energy,* **16(6)**. DOI: 10.1063/5.0222974

[9] Reddy, P. J., Chinta, S., **Baki, H.,** Matear, R., & Taylor, J. (2024). Gaussian process regression-based Bayesian Optimisation (G-BO) of model parameters—a WRF model case study of southeast Australia heat extremes. *Geophysical Research Letters*, ***51*(17), e2024GL111074**. DOI: 10.1029/2024GL111074

[8] **Baki, H.,** & Basu, S. (2024). Estimating high-resolution profiles of wind speeds from a global reanalysis dataset using TabNet. *Environmental Data Science,* **2024;3:e32**. DOI: 10.1017/eds.2024.41

[7] Reddy, P. J., **Baki, H.,** Chinta, S., Matear, R., & Taylor, J. (2023). PAUNet: Precipitation Attention-based U-Net for rain prediction from satellite radiance data. *arXiv preprint* DOI: arXiv:2311.18306

[6] Reddy, P. J., Chinta, S., Matear, R., Taylor, J., **Baki, H.,** Thatcher, M., ... & Sharples, J. (2023). Machine learning based parameter sensitivity of regional climate models—a case study of the WRF model for heat extremes over Southeast Australia. *Environmental research letters*, ***19*(1), 014010**. DOI: 10.1088/1748-9326/ad0eb0

[5] **Baki, H.,** Chinta, S., Balaji, C., and Srinivasan,B. (2022). Determining the sensitive parameters of the Weather Research and Forecasting (WRF) model for the simulation of tropical cyclones in the Bay of Bengal using global sensitivity analysis and machine learning, *Geoscientific Model Development,* **15(5), pp.2133-2155**. DOI: 10.5194/gmd-15-2133-2022

[4] **Baki, H.,** Chinta, S., Balaji,C., and Srinivasan,B. (2022). WRF model parameter calibration to improve the prediction of tropical cyclones over the Bay of Bengal using Machine Learning-based Mult-objective Optimization, *Journal of Applied Meteorology and Climatology, **61(7), pp.819-83***. https://doi.org/10.1175/JAMC-D-21-0184.1

[3] **Baki, H.,** Balaji, C., and Srinivasan, B. (2022). Impact of data assimilation on a calibrated WRF model for the prediction of tropical cyclones over the Bay of Bengal, *Current Science*, **122(5), pp.569-583**. https://doi.org/10.18520/cs/v122/i5/569-583

[2] **Baki, H.,** Chinta, S., Balaji, C., and Srinivasan, B. (2021). A sensitivity study of WRF model microphysics and cumulus parameterization schemes for the simulations of tropical cyclones using GPM radar data, *Journal of Earth System Science* **130(4), pp.1-30**. https://doi.org/10.1007/s12040-021-01682-3

[1] **Baki, H.,** Kumar, K. E. S., & Srinivasan, B. (2020). Topology Optimization Using Convolutional Neural Network. In *Advances in Multidisciplinary Analysis and Optimization* **(pp.301-307)**. Springer, Singapore. https://doi.org/10.1007/978-981-15-5432-2_26

BOOKS

[1] Chinta, S., **Baki, H.,** Balaji, C., and Srinivasan, B. (2024). Fine-Tuning Extreme Rainfall Predictions: A Machine Learning Approach - Sandeep Chinta, Harish Baki, C. Balaji & Balaji Srinivasan, *Ane Books Pvt. Ltd.* ISBN: 9788119662289

CONFERENCE PRESENTATIONS

[10] **Baki, H.,** Maximilian, P., Basu, S., (2025, Jun). S2GDI: A Unified Deep Learning Framework for Reconstructing Continuous Gridded Meteorological Fields from Sparse Station Observations.

Gordon Research Conference - Machine Learning for Actionable Climate Science, June 22-27, 2025.

[9] **Baki, H.,** Basu, S., (2025, Jun). DAMAGE - A **D**eep Learning **A**pproach for **M**eteorological **A**nalysis & **G**ust **E**stimation from sparse station observation. Annual WISER IAB Meeting, June 12-13, 2025.

[8] **Baki, H.,** Basu, S., (2025, Jan). Modeling Frontal Low-Level Jets and Associated Extreme Wind Ramps. 105th Annual AMS Meeting 2025 105, 452764.

[7] **Baki, H.,** Basu, S., (2025, Jan). Estimating Offshore Wind Power Potential by Utilizing Gray-Zone Atmospheric Modeling. 105th Annual AMS Meeting 2025 105, 453681.

[6] **Baki, H.,** Basu, S., (2025, Jan). Estimating High-Resolution Wind Speed Profiles from a Reanalysis Dataset using TabNet: A Transfer Learning Approach. 105th Annual AMS Meeting 2025 105, 453572.

[5] **Baki, H.,** Basu, S., (2024)**.** A Chebyshev Polynomial-based Wind Speed Profile Characterization Framework: Applications in Mesoscale Model Evaluation. NAWEA-WindTech 2024, October 30-November 1, 2024, Hyatt Regency, New Brunswick, NJ.

[4] **Baki, H.,** & Basu, S., (2024, April). Estimating high-resolution profiles of wind speeds from a global reanalysis dataset using TabNet. Climate Informatics 2024.

[3] **Baki, H.,** Basu, S., & Lavidas, G., (2023, May). Statistical characterization of simulated wind ramps. In *EGU General Assembly Conference Abstracts* (pp. EGU-17208).

[2] **Baki, H.,** Chinta, S., Balaji, C., and Srinivasan, B. (2021). Use of Machine Learning algorithms in evaluating the WRF model parameter sensitivity for the simulations of tropical cyclones, *vEGU21, the 23rd EGU General Assembly*, held online 19-30 April 2021, id. EGU21-5826.

[1] **Baki, H.,** Chinta, S., Balaji, C., and Srinivasan, B. (2021). A Preliminary Study of GPM Radar Reflectivity Assimilation using WRF model for Tropical Cyclones, *4th National Conference on India Radar Meteorology*, 5-7 February 2020, IIT Madras, India.

REVIEWING ACTIVITIES

Papers reviewed for: Environmental Research Letter; Journal of Hydrology, wind energy science

TEACHING

Delft University of Technology

- CIEM4210: Marine Renewables (Module B1, Q4, 2024)
- Lecture 11 on Wind resource assessment

RESEARCH MENTORSHIP

Dr. Jaya Singh (Postdoctoral associate, Atmospheric Sciences Research Center, UAlbany)

- Provided technical training on the application of machine learning for wind gust estimation from global reanalysis datasets

Ekaterina Belash (Graduate research assistant, Atmospheric Sciences Research Center, UAlbany)

- Conducted hands-on training on NVIDIA A100 GPU resources, focusing on multi-GPU execution and development of deep-learning frameworks for wind-gust downscaling

Suqian Chu (Graduate research assistant, Atmospheric Sciences Research Center, UAlbany)

- Provided hands-on training on NVIDIA A100 GPU resources with emphasis on implementing deep-learning-based downscaling for future climate projections